

SEC P vs NP — SUPPORTED findings

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 16:26 UTC

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□ **AUDIT 2026-05-08:** this report has been filtered against `retractions.json`. Some entries previously listed here have been retracted following a code-level audit. See [AUDIT_2026-05-08.md](#) for the full audit document and [MULTIAGENT_PIPELINE.md](#) for the new review pipeline.

Compiled 2026-05-18 16:26 UTC from `pvsnp_notebook.jsonl`. 0 conjectures empirically supported (on small instances; all require follow-up at larger n).

Important caveat: these are *empirical* results on instances of size ≤ 20 . A SUPPORTED verdict here means the test did not find a counterexample in the sampled regime. Genuine mathematical validation requires: (i) extending the test to $n \geq 50$, (ii) proving the bound analytically, (iii) independent reproduction.

No SUPPORTED verdicts in the curated set.

Retractions (originally SUPPORTED)

The following 3 entries were removed from this report on 2026-05-08 per the audit document. They are preserved in the raw notebook/`*.jsonl` for traceability but are NOT to be cited as scientific output.

7cbbaa3e1e4a — Tropical Rank of Clause-Indicator Polynomial Bounds ACC Circuit Size

- **Original verdict:** SUPPORTED
- **Action:** RETRACTED
- **Reason:** Malformed test: `cnf_to_tropical_matrix` uses `len(cnf[0])` as variable count; `acc_circuit_size` counts clauses+literals, no relation to ACC^0 complexity. The reported `support_fraction` is an XNOR artefact in the chosen n . Novelty filter recorded 0 arXiv hits.

b43a4129e5c5 — Ideal Generators Count Bounds Communication Complexity

- **Original verdict:** SUPPORTED
- **Action:** RETRACTED
- **Reason:** Pure stub: `comm_complexity(f, n) := n`, `ideal_generators(f, n) := 2*n`. Test reduces to $n \leq 2n$, trivially true. Output line contained the unresolved placeholder 'RESULT: SUPPORTED ='. Novelty filter recorded 0 arXiv hits.

e006a48b37a7 — Frobenius-Schur Indicator of Clause-Symmetry Group Predicts SAT Symmetry Breaking Cost

- **Original verdict:** SUPPORTED
- **Action:** RETRACTED

- **Reason:** Stub test: frobenius_schur_sum returns hard-coded constants based on string label; add_lex_leader_clause returns a fixed number of dummy clauses. Reported ratio_avg is the deterministic average of those constants. No CDCL run, no representation theory computed.
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Demoted entries (originally SUPPORTED, now INCONCLUSIVE pending reformulation)

15ae8fd62af0 — Grothendieck-Witt Class of Clause Polynomials Mod 2 Predicts Resolution Width

- **Original verdict:** SUPPORTED
- **New verdict:** INCONCLUSIVE
- **Action:** DEMOTED_PENDING_REFORMULATION
- **Reason:** Statement-test mismatch. The conjecture is stated over F_2 where $Q(x) = \sum \langle c, x \rangle^2 \bmod 2$ is linear (since $a^2 = a$ in F_2), making the bilinear form B identically zero and the conjecture vacuous. The test silently substitutes ‘integer inner product, then squared, then reduced mod 2’, i.e. tests a different statement. The underlying intuition (Grothendieck-Witt class of clause polynomials) may be salvageable via re-formulation with an explicit ring assignment; pending such re-formulation, demoted to INCONCLUSIVE.